

Lexical Grammar

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1 Regex

1.1 Substitutions

$$\begin{array}{lll}
L & := & [\mathbf{a} - \mathbf{z}] \quad (1) \\
U & := & [\mathbf{A} - \mathbf{Z}] \quad (2) \\
D & := & [0 - 9] \quad (3) \\
H & := & [\mathbf{a} - \mathbf{fA} - \mathbf{F0} - 9] \quad (4) \\
O & := & [0 - 7] \quad (5) \\
B & := & [01] \quad (6)
\end{array}$$

1.2 Identifiers, Constants, and Comments

-{0, 2}L(? : L U D _)*	→	IDENT		(7)
U(? : L U D)*	→	TYPE_IDENT		(8)
[\ + -]? D +	→	NUMERIC_CONST	Integer	(9)
0[xX]H {1, 16}	→	NUMERIC_CONST	Hex	(10)
0[oO]O {1, 32}	→	NUMERIC_CONST	Octal	(11)
0[bB]B {1, 64}	→	NUMERIC_CONST	Binary	(12)
[\ + -]? D + [eE][\ + -]? D +	→	NUMERIC_CONST	Scientific Notation	(13)
[\ + -]?(? : D + (? : [.] D *)?[.] D +)	→	NUMERIC_CONST	Float	(14)
r?"((? : [^ " \\ \\\ \\\ ")*)"	→	STRING_LITERAL		(15)
#. * \n\$	→	COMMENT		(16)

1.3 Symbols

,	→	COMMA	(17)
;	→	SEMI	(18)
(	→	LEFT_PAREN	(19)
)	→	RIGHT_PAREN	(20)
[	→	LEFT_SQUARE	(21)
]	→	RIGHT_SQUARE	(22)
{	→	LEFT_CURLY	(23)
}	→	RIGHT_CURLY	(24)

1.4 Keywords

and	→	LOGICAL_AND	(25)
bool	→	BOOL	(26)
break	→	BREAK	(27)
class	→	CLASS	(28)
elif	→	ELIF	(29)
else	→	ELSE	(30)
enum	→	ENUM	(31)
exit	→	EXIT	(32)
false	→	FALSE	(33)
float32	→	FLOAT32	(34)
float64	→	FLOAT64	(35)
for	→	FOR	(36)
func	→	FUNCTION	(37)
if	→	IF	(38)
iface	→	INTERFACE	(39)
in	→	IN	(40)
int16	→	INT16	(41)
int32	→	INT32	(42)
int64	→	INT64	(43)
int8	→	INT8	(44)
mut	→	MUTABLE	(45)
or	→	LOGICAL_OR	(46)
private	→	PRIVATE	(47)
public	→	PUBLIC	(48)
string	→	STRING	(49)
true	→	TRUE	(50)
uint16	→	UINT16	(51)
uint32	→	UINT32	(52)
uint64	→	UINT64	(53)
uint8	→	UINT8	(54)
while	→	WHILE	(55)

1.5 Operators

$\wedge\wedge=$	→	EXPONENT_ASSIGN	(56)
$\wedge\wedge$	→	EXPONENT	(57)
$\wedge=$	→	XOR_ASSIGN	(58)
$\wedge$	→	XOR	(59)
$\% =$	→	MOD_ASSIGN	(60)
$\%$	→	MOD	(61)
$* =$	→	MUL_ASSIGN	(62)
$*$	→	MUL	(63)
$// =$	→	INTEGER_DIV_ASSIGN	(64)
$//$	→	INTEGER_DIV	(65)
$/ =$	→	DIV_ASSIGN	(66)
$/$	→	DIV	(67)
$++$	→	INC	(68)
$+ =$	→	ADD_ASSIGN	(69)
$+$	→	ADD	(70)
$--$	→	DEC	(71)
$- =$	→	SUB_ASSIGN	(72)
$-$	→	SUB	(73)
$<<$	→	SHIFT_LEFT	(74)
$<=$	→	LE_OP	(75)
$<$	→	LT_OP	(76)
$>>$	→	SHIFT_RIGHT	(77)
$>=$	→	GE_OP	(78)
$>$	→	GT_OP	(79)
$!=$	→	NE_OP	(80)
$==$	→	EQ_OP	(81)
$=$	→	ASSIGN	(82)
$\& =$	→	AND_ASSIGN	(83)
$\&$	→	AND	(84)
$ =$	→	OR_ASSIGN	(85)
$ $	→	OR	(86)
$?$	→	TERNARY	(87)
$.$	→	DOT	(88)

2 Backus-Naur Form

$\langle \text{ident} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(89)
		$ \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(90)
		$ \text{"_"} \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(91)
$\langle \text{ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(92)
		$ \langle \text{upper} \rangle \langle \text{ident?} \rangle$	(93)
		$ \langle \text{digit} \rangle \langle \text{ident?} \rangle$	(94)
		$ \text{"_"} \langle \text{ident?} \rangle$	(95)
		$ \varepsilon$	(96)
$\langle \text{type_ident} \rangle$	$::=$	$\langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(97)
$\langle \text{type_ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{type_ident?} \rangle$	(98)
		$ \langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(99)
		$ \langle \text{digit} \rangle \langle \text{type_ident?} \rangle$	(100)
		$ \varepsilon$	(101)
$\langle \text{float_const} \rangle$	$::=$	$\langle \text{mantissa} \rangle$	(102)
		$ \langle \text{hex_const} \rangle$	(103)
		$ \langle \text{octal_const} \rangle$	(104)
		$ \langle \text{binary_const} \rangle$	(105)
		$ \langle \text{scientific} \rangle$	(106)
$\langle \text{integer_const} \rangle$	$::=$	$\langle \text{decimal_const} \rangle$	(107)
		$ \langle \text{hex_const} \rangle$	(108)
		$ \langle \text{octal_const} \rangle$	(109)
		$ \langle \text{binary_const} \rangle$	(110)
$\langle \text{comment} \rangle$	$::=$	$\text{"\#"} \langle \text{comment_text} \rangle$	(111)
$\langle \text{comment_text} \rangle$	$::=$	$\langle \text{comment_char} \rangle \langle \text{comment_text} \rangle$	(112)
$\langle \text{comment_char} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{upper} \rangle \langle \text{digit} \rangle \langle \text{symbol} \rangle \langle \text{non_newline} \rangle$	(113)
$\langle \text{scientific} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{mantissa} \rangle \langle \text{exponent} \rangle$	(114)
$\langle \text{mantissa} \rangle$	$::=$	$\langle \text{digits} \rangle \text{"."} \langle \text{digits?} \rangle$	(115)
		$ \text{"."} \langle \text{digits} \rangle$	(116)
		$ \langle \text{digits} \rangle$	(117)
$\langle \text{exponent} \rangle$	$::=$	$\text{"e"} \langle \text{decimal_const} \rangle$	(118)
$\langle \text{decimal_const} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{digits} \rangle$	(119)
$\langle \text{hex_const} \rangle$	$::=$	$\text{"0"} \text{"x"} \langle \text{hs} \rangle$	(120)
		$ \text{"0"} \text{"X"} \langle \text{hs} \rangle$	(121)
$\langle \text{octal_const} \rangle$	$::=$	$\text{"0"} \text{"o"} \langle \text{os} \rangle$	(122)
		$ \text{"0"} \text{"O"} \langle \text{os} \rangle$	(123)
$\langle \text{binary_const} \rangle$	$::=$	$\text{"0"} \text{"b"} \langle \text{bs} \rangle$	(124)
		$ \text{"0"} \text{"B"} \langle \text{bs} \rangle$	(125)
$\langle \text{digits?} \rangle$	$::=$	$\langle \text{digits} \rangle \varepsilon$	(126)
$\langle \text{digits} \rangle$	$::=$	$\langle \text{digit} \rangle \langle \text{digit} \rangle \langle \text{digits} \rangle$	(127)
$\langle \text{sign?} \rangle$	$::=$	$\langle \text{sign} \rangle \varepsilon$	(128)
$\langle \text{sign} \rangle$	$::=$	$\text{"+"} \text{"-"} \quad 4$	(129)
			(130)

$\langle lower \rangle ::= "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h" | "i" | "j" | "k" | "l" | "m" |$
 $"n" | "o" | "p" | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x" | "y" | "z"$ (131)

$\langle upper \rangle ::= "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H" | "I" | "J" | "K" | "L" | "M" |$
 $"N" | "O" | "P" | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X" | "Y" | "Z"$ (132)

$\langle digit \rangle ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"$ (133)

$\langle hex \rangle ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9" |$
 $"a" | "b" | "c" | "d" | "e" | "f" | "A" | "B" | "C" | "D" | "E" | "F"$ (134)

$\langle octal \rangle ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7"$ (135)

$\langle binary \rangle ::= "0" | "1"$ (136)

$\langle symbols \rangle ::= "!" | "\"" | "\#" | "$" | "%" | "&" | "'" | "(" | ")" | "*" | "+" | "," | "-" | "." | "/" | ":" | ";" |$
 $"<" | "=" | ">" | "?" | "@" | "[" | "]" | "\" | "]" | "_" | "`" | "{" | "}" | "$ (137)

$\langle whitespace \rangle ::= \langle non_newline \rangle | \langle newline \rangle$ (138)

$\langle non_newline \rangle ::= " " | "\t" | "\r"$ (139)

$\langle newline \rangle ::= "\n"$ (140)

$\langle hs \rangle$	$::=$	$\langle hex \rangle$	(141)
		$ \langle hex \rangle \langle hs1 \rangle$	(142)
$\langle hs1 \rangle$	$::=$	$\langle hex \rangle$	(143)
		$ \langle hex \rangle \langle hs2 \rangle$	(144)
$\langle hs2 \rangle$	$::=$	$\langle hex \rangle$	(145)
		$ \langle hex \rangle \langle hs3 \rangle$	(146)
$\langle hs3 \rangle$	$::=$	$\langle hex \rangle$	(147)
		$ \langle hex \rangle \langle hs4 \rangle$	(148)
$\langle hs4 \rangle$	$::=$	$\langle hex \rangle$	(149)
		$ \langle hex \rangle \langle hs5 \rangle$	(150)
$\langle hs5 \rangle$	$::=$	$\langle hex \rangle$	(151)
		$ \langle hex \rangle \langle hs6 \rangle$	(152)
$\langle hs6 \rangle$	$::=$	$\langle hex \rangle$	(153)
		$ \langle hex \rangle \langle hs7 \rangle$	(154)
$\langle hs7 \rangle$	$::=$	$\langle hex \rangle$	(155)
		$ \langle hex \rangle \langle hs8 \rangle$	(156)
$\langle hs8 \rangle$	$::=$	$\langle hex \rangle$	(157)
		$ \langle hex \rangle \langle hs9 \rangle$	(158)
$\langle hs9 \rangle$	$::=$	$\langle hex \rangle$	(159)
		$ \langle hex \rangle \langle hs10 \rangle$	(160)
$\langle hs10 \rangle$	$::=$	$\langle hex \rangle$	(161)
		$ \langle hex \rangle \langle hs11 \rangle$	(162)
$\langle hs11 \rangle$	$::=$	$\langle hex \rangle$	(163)
		$ \langle hex \rangle \langle hs12 \rangle$	(164)
$\langle hs12 \rangle$	$::=$	$\langle hex \rangle$	(165)
		$ \langle hex \rangle \langle hs13 \rangle$	(166)
$\langle hs13 \rangle$	$::=$	$\langle hex \rangle$	(167)
		$ \langle hex \rangle \langle hs14 \rangle$	(168)
$\langle hs14 \rangle$	$::=$	$\langle hex \rangle$	(169)
		$ \langle hex \rangle \langle hs15 \rangle$	(170)
$\langle hs15 \rangle$	$::=$	$\langle hex \rangle$	(171)

$\langle os \rangle$	$::=$	$\langle octal \rangle$	(172)
		$ \langle octal \rangle \langle os1 \rangle$	(173)
$\langle os1 \rangle$	$::=$	$\langle octal \rangle$	(174)
		$ \langle octal \rangle \langle os2 \rangle$	(175)
$\langle os2 \rangle$	$::=$	$\langle octal \rangle$	(176)
		$ \langle octal \rangle \langle os3 \rangle$	(177)
$\langle os3 \rangle$	$::=$	$\langle octal \rangle$	(178)
		$ \langle octal \rangle \langle os4 \rangle$	(179)
$\langle os4 \rangle$	$::=$	$\langle octal \rangle$	(180)
		$ \langle octal \rangle \langle os5 \rangle$	(181)
$\langle os5 \rangle$	$::=$	$\langle octal \rangle$	(182)
		$ \langle octal \rangle \langle os6 \rangle$	(183)
$\langle os6 \rangle$	$::=$	$\langle octal \rangle$	(184)
		$ \langle octal \rangle \langle os7 \rangle$	(185)
$\langle os7 \rangle$	$::=$	$\langle octal \rangle$	(186)
		$ \langle octal \rangle \langle os8 \rangle$	(187)
$\langle os8 \rangle$	$::=$	$\langle octal \rangle$	(188)
		$ \langle octal \rangle \langle os9 \rangle$	(189)
$\langle os9 \rangle$	$::=$	$\langle octal \rangle$	(190)
		$ \langle octal \rangle \langle os10 \rangle$	(191)
$\langle os10 \rangle$	$::=$	$\langle octal \rangle$	(192)
		$ \langle octal \rangle \langle os11 \rangle$	(193)
$\langle os11 \rangle$	$::=$	$\langle octal \rangle$	(194)
		$ \langle octal \rangle \langle os12 \rangle$	(195)
$\langle os12 \rangle$	$::=$	$\langle octal \rangle$	(196)
		$ \langle octal \rangle \langle os13 \rangle$	(197)
$\langle os13 \rangle$	$::=$	$\langle octal \rangle$	(198)
		$ \langle octal \rangle \langle os14 \rangle$	(199)
$\langle os14 \rangle$	$::=$	$\langle octal \rangle$	(200)
		$ \langle octal \rangle \langle os15 \rangle$	(201)
$\langle os15 \rangle$	$::=$	$\langle octal \rangle$	(202)
		$ \langle octal \rangle \langle os16 \rangle$	(203)
$\langle os16 \rangle$	$::=$	$\langle octal \rangle$	(204)
		$ \langle octal \rangle \langle os17 \rangle$	(205)
$\langle os17 \rangle$	$::=$	$\langle octal \rangle$	(206)
		$ \langle octal \rangle \langle os18 \rangle$	(207)
$\langle os18 \rangle$	$::=$	$\langle octal \rangle$	(208)
		$ \langle octal \rangle \langle os19 \rangle$	(209)
$\langle os19 \rangle$	$::=$	$\langle octal \rangle$	(210)
		$ \langle octal \rangle \langle os20 \rangle$	(211)

$\langle os20 \rangle$	$::=$	$\langle octal \rangle$	(212)
		$ \langle octal \rangle \langle os21 \rangle$	(213)
$\langle os21 \rangle$	$::=$	$\langle octal \rangle$	(214)
		$ \langle octal \rangle \langle os22 \rangle$	(215)
$\langle os22 \rangle$	$::=$	$\langle octal \rangle$	(216)
		$ \langle octal \rangle \langle os23 \rangle$	(217)
$\langle os23 \rangle$	$::=$	$\langle octal \rangle$	(218)
		$ \langle octal \rangle \langle os24 \rangle$	(219)
$\langle os24 \rangle$	$::=$	$\langle octal \rangle$	(220)
		$ \langle octal \rangle \langle os25 \rangle$	(221)
$\langle os25 \rangle$	$::=$	$\langle octal \rangle$	(222)
		$ \langle octal \rangle \langle os26 \rangle$	(223)
$\langle os26 \rangle$	$::=$	$\langle octal \rangle$	(224)
		$ \langle octal \rangle \langle os27 \rangle$	(225)
$\langle os27 \rangle$	$::=$	$\langle octal \rangle$	(226)
		$ \langle octal \rangle \langle os28 \rangle$	(227)
$\langle os28 \rangle$	$::=$	$\langle octal \rangle$	(228)
		$ \langle octal \rangle \langle os29 \rangle$	(229)
$\langle os29 \rangle$	$::=$	$\langle octal \rangle$	(230)
		$ \langle octal \rangle \langle os30 \rangle$	(231)
$\langle os30 \rangle$	$::=$	$\langle octal \rangle$	(232)
		$ \langle octal \rangle \langle os31 \rangle$	(233)
$\langle os31 \rangle$	$::=$	$\langle octal \rangle$	(234)
$\langle bs \rangle$	$::=$	$\langle binary \rangle$	(235)
		$ \langle binary \rangle \langle bs1 \rangle$	(236)
$\langle bs1 \rangle$	$::=$	$\langle binary \rangle$	(237)
		$ \langle binary \rangle \langle bs2 \rangle$	(238)
$\langle bs2 \rangle$	$::=$	$\langle binary \rangle$	(239)
		$ \langle binary \rangle \langle bs3 \rangle$	(240)
$\langle bs3 \rangle$	$::=$	$\langle binary \rangle$	(241)
		$ \langle binary \rangle \langle bs4 \rangle$	(242)
$\langle bs4 \rangle$	$::=$	$\langle binary \rangle$	(243)
		$ \langle binary \rangle \langle bs5 \rangle$	(244)
$\langle bs5 \rangle$	$::=$	$\langle binary \rangle$	(245)
		$ \langle binary \rangle \langle bs6 \rangle$	(246)
$\langle bs6 \rangle$	$::=$	$\langle binary \rangle$	(247)
		$ \langle binary \rangle \langle bs7 \rangle$	(248)

$\langle bs7 \rangle$	::=	$\langle binary \rangle$	(249)
		$ \langle binary \rangle \langle bs8 \rangle$	(250)
$\langle bs8 \rangle$	::=	$\langle binary \rangle$	(251)
		$ \langle binary \rangle \langle bs9 \rangle$	(252)
$\langle bs9 \rangle$	::=	$\langle binary \rangle$	(253)
		$ \langle binary \rangle \langle bs10 \rangle$	(254)
$\langle bs10 \rangle$	::=	$\langle binary \rangle$	(255)
		$ \langle binary \rangle \langle bs11 \rangle$	(256)
$\langle bs11 \rangle$	::=	$\langle binary \rangle$	(257)
		$ \langle binary \rangle \langle bs12 \rangle$	(258)
$\langle bs12 \rangle$	::=	$\langle binary \rangle$	(259)
		$ \langle binary \rangle \langle bs13 \rangle$	(260)
$\langle bs13 \rangle$	::=	$\langle binary \rangle$	(261)
		$ \langle binary \rangle \langle bs14 \rangle$	(262)
$\langle bs14 \rangle$	::=	$\langle binary \rangle$	(263)
		$ \langle binary \rangle \langle bs15 \rangle$	(264)
$\langle bs15 \rangle$	::=	$\langle binary \rangle$	(265)
		$ \langle binary \rangle \langle bs16 \rangle$	(266)
$\langle bs16 \rangle$	::=	$\langle binary \rangle$	(267)
		$ \langle binary \rangle \langle bs17 \rangle$	(268)
$\langle bs17 \rangle$	::=	$\langle binary \rangle$	(269)
		$ \langle binary \rangle \langle bs18 \rangle$	(270)
$\langle bs18 \rangle$	::=	$\langle binary \rangle$	(271)
		$ \langle binary \rangle \langle bs19 \rangle$	(272)
$\langle bs19 \rangle$	::=	$\langle binary \rangle$	(273)
		$ \langle binary \rangle \langle bs20 \rangle$	(274)
$\langle bs20 \rangle$	::=	$\langle binary \rangle$	(275)
		$ \langle binary \rangle \langle bs21 \rangle$	(276)
$\langle bs21 \rangle$	::=	$\langle binary \rangle$	(277)
		$ \langle binary \rangle \langle bs22 \rangle$	(278)
$\langle bs22 \rangle$	::=	$\langle binary \rangle$	(279)
		$ \langle binary \rangle \langle bs23 \rangle$	(280)
$\langle bs23 \rangle$	::=	$\langle binary \rangle$	(281)
		$ \langle binary \rangle \langle bs24 \rangle$	(282)
$\langle bs24 \rangle$	::=	$\langle binary \rangle$	(283)
		$ \langle binary \rangle \langle bs25 \rangle$	(284)
$\langle bs25 \rangle$	::=	$\langle binary \rangle$	(285)
		$ \langle binary \rangle \langle bs26 \rangle$	(286)
$\langle bs26 \rangle$	::=	$\langle binary \rangle$	(287)
		$ \langle binary \rangle \langle bs27 \rangle$	(288)

$\langle bs27 \rangle$	::=	$\langle binary \rangle$	(289)
		$ \langle binary \rangle \langle bs28 \rangle$	(290)
$\langle bs28 \rangle$	::=	$\langle binary \rangle$	(291)
		$ \langle binary \rangle \langle bs29 \rangle$	(292)
$\langle bs29 \rangle$	::=	$\langle binary \rangle$	(293)
		$ \langle binary \rangle \langle bs30 \rangle$	(294)
$\langle bs30 \rangle$	::=	$\langle binary \rangle$	(295)
		$ \langle binary \rangle \langle bs31 \rangle$	(296)
$\langle bs31 \rangle$	::=	$\langle binary \rangle$	(297)
		$ \langle binary \rangle \langle bs32 \rangle$	(298)
$\langle bs32 \rangle$	::=	$\langle binary \rangle$	(299)
		$ \langle binary \rangle \langle bs33 \rangle$	(300)
$\langle bs33 \rangle$	::=	$\langle binary \rangle$	(301)
		$ \langle binary \rangle \langle bs34 \rangle$	(302)
$\langle bs34 \rangle$	::=	$\langle binary \rangle$	(303)
		$ \langle binary \rangle \langle bs35 \rangle$	(304)
$\langle bs35 \rangle$	::=	$\langle binary \rangle$	(305)
		$ \langle binary \rangle \langle bs36 \rangle$	(306)
$\langle bs36 \rangle$	::=	$\langle binary \rangle$	(307)
		$ \langle binary \rangle \langle bs37 \rangle$	(308)
$\langle bs37 \rangle$	::=	$\langle binary \rangle$	(309)
		$ \langle binary \rangle \langle bs38 \rangle$	(310)
$\langle bs38 \rangle$	::=	$\langle binary \rangle$	(311)
		$ \langle binary \rangle \langle bs39 \rangle$	(312)
$\langle bs39 \rangle$	::=	$\langle binary \rangle$	(313)
		$ \langle binary \rangle \langle bs40 \rangle$	(314)
$\langle bs40 \rangle$	::=	$\langle binary \rangle$	(315)
		$ \langle binary \rangle \langle bs41 \rangle$	(316)
$\langle bs41 \rangle$	::=	$\langle binary \rangle$	(317)
		$ \langle binary \rangle \langle bs42 \rangle$	(318)
$\langle bs42 \rangle$	::=	$\langle binary \rangle$	(319)
		$ \langle binary \rangle \langle bs43 \rangle$	(320)
$\langle bs43 \rangle$	::=	$\langle binary \rangle$	(321)
		$ \langle binary \rangle \langle bs44 \rangle$	(322)
$\langle bs44 \rangle$	::=	$\langle binary \rangle$	(323)
		$ \langle binary \rangle \langle bs45 \rangle$	(324)
$\langle bs45 \rangle$	::=	$\langle binary \rangle$	(325)
		$ \langle binary \rangle \langle bs46 \rangle$	(326)
$\langle bs46 \rangle$	::=	$\langle binary \rangle$	(327)
		$ \langle binary \rangle \langle bs47 \rangle$	(328)

$\langle bs47 \rangle$	$::=$	$\langle binary \rangle$	(329)
		$ \langle binary \rangle \langle bs48 \rangle$	(330)
$\langle bs48 \rangle$	$::=$	$\langle binary \rangle$	(331)
		$ \langle binary \rangle \langle bs49 \rangle$	(332)
$\langle bs49 \rangle$	$::=$	$\langle binary \rangle$	(333)
		$ \langle binary \rangle \langle bs50 \rangle$	(334)
$\langle bs50 \rangle$	$::=$	$\langle binary \rangle$	(335)
		$ \langle binary \rangle \langle bs51 \rangle$	(336)
$\langle bs51 \rangle$	$::=$	$\langle binary \rangle$	(337)
		$ \langle binary \rangle \langle bs52 \rangle$	(338)
$\langle bs52 \rangle$	$::=$	$\langle binary \rangle$	(339)
		$ \langle binary \rangle \langle bs53 \rangle$	(340)
$\langle bs53 \rangle$	$::=$	$\langle binary \rangle$	(341)
		$ \langle binary \rangle \langle bs54 \rangle$	(342)
$\langle bs54 \rangle$	$::=$	$\langle binary \rangle$	(343)
		$ \langle binary \rangle \langle bs55 \rangle$	(344)
$\langle bs55 \rangle$	$::=$	$\langle binary \rangle$	(345)
		$ \langle binary \rangle \langle bs56 \rangle$	(346)
$\langle bs56 \rangle$	$::=$	$\langle binary \rangle$	(347)
		$ \langle binary \rangle \langle bs57 \rangle$	(348)
$\langle bs57 \rangle$	$::=$	$\langle binary \rangle$	(349)
		$ \langle binary \rangle \langle bs58 \rangle$	(350)
$\langle bs58 \rangle$	$::=$	$\langle binary \rangle$	(351)
		$ \langle binary \rangle \langle bs59 \rangle$	(352)
$\langle bs59 \rangle$	$::=$	$\langle binary \rangle$	(353)
		$ \langle binary \rangle \langle bs60 \rangle$	(354)
$\langle bs60 \rangle$	$::=$	$\langle binary \rangle$	(355)
		$ \langle binary \rangle \langle bs61 \rangle$	(356)
$\langle bs61 \rangle$	$::=$	$\langle binary \rangle$	(357)
		$ \langle binary \rangle \langle bs62 \rangle$	(358)
$\langle bs62 \rangle$	$::=$	$\langle binary \rangle$	(359)
		$ \langle binary \rangle \langle bs63 \rangle$	(360)
$\langle bs63 \rangle$	$::=$	$\langle binary \rangle$	(361)
			(362)